# Practical 2



**Electricity and Electronics EBN 111**

Semester 1, 2024

| Practical Information | | | |
|---|---|---|---|
| Maximum marks: | 150 | Full marks: | 150 |
| Duration of practical: | 180 min | Open/Closed book: | Open |
| Total number of pages (this page included) | | 16 | |

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**Assistant Lecturers:** S. Twala, L. Hurter, J. Masela, J Nepembe

## Overview of Practicals

The engineering process to build any circuit consists of three parts:

- **Calculations**
  A circuit design is proposed. The circuit is analyzed using fundamental circuit theory, e.g., Ohm's Law, Kirchoff's Laws, Nodal analysis, Mesh analysis etc. Calculations are done by hand to determine specific voltages and/or currents in the circuit. This is the first step to ensure the proposed circuit design is correct.
- **Simulation**
  The proposed circuit is simulated using software such as LTspice. The simulation includes non-ideal effects which are not included in the calculations. The simulation results should coincide closely with the calculations done previously. The simulation allows us to analyze the circuit for a wide variety of circumstances. The simulations verify that the proposed circuit will perform its intended function.
- **Laboratory/Practical**
  The proposed circuit is built using physical components. The hardware is connected to an actual power source such as a battery. Laboratory equipment can be used to measure the voltages or currents determined in the calculation and simulation steps above. The measurements should coincide closely with the results in the *Calculation* step. It will not be exact as there are tolerances in the physical components which are not included in the *Calculations*. The measurements should also be similar to the simulations, but will not be exactly the same since not all non-ideal characteristics are included in the simulation.

## Practicals Process

The EBN 111 Practicals will follow the engineering process discussed above. As shown in the figure below, a complete Practical consists of a **Pre-Practical**, a **Demonstration**, and a **Laboratory** part. The Laboratory part is also often referred to as a ‘Practical session’.

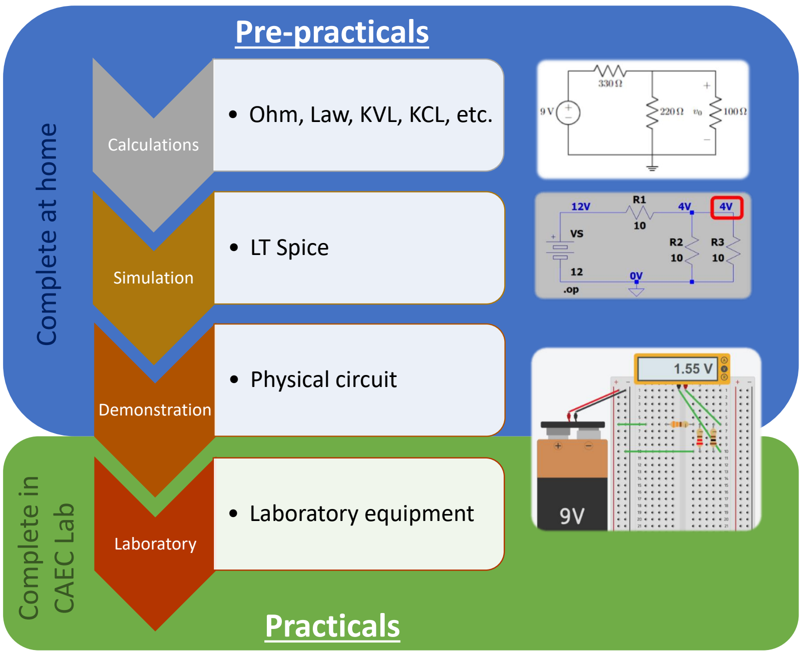
1. **Section 1: Pre-Practical - Calculations and Simulation** [**60 marks**]
    - Analyse the circuit diagrams in the practical guide using the theory covered in EBN 111. *Calculate* the various unknowns.
    - *Simulate* the circuits using LTspice and compare the results to your calculations.
    - Download the Excel spreadsheet for your Pre-Practical on AMS and fill in your answers. Submit your Excel spreadsheet on AMS before the relevant deadline.
    - You will have to work through the guideline documentation to familiarise yourself with the circuit components and software that are required for the pre-practical.

**Pre-Practicals are completed at home before a Practical session.**

2. **Section 2: Practical Demonstration** [**20 marks**]
    - You will be given a circuit similar to a circuit you analyzed and simulated to build on a breadboard using the component kit provided for EBN 111. You will demonstrate this physical circuit upon entering the laboratory. We will award a mark for the circuit.

3. **Section 3: Practical Measurements** [**60 marks**]
    - Use laboratory equipment to measure voltages and currents as specified in the practical guide for the circuit you built. Compare it to your Pre-Practical results.
    - Download the Excel spreadsheet for your Practical on AMS and fill in your answers. Submit your Excel spreadsheet on AMS before the end of the practical session.

**Section 3 is completed in the CAEC Lab during a 3-hour Practical session.**
**All measurements must be made and submitted during the Practical session.**



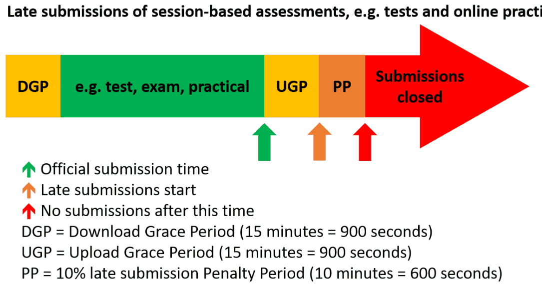
## Assessment evaluation, submission rules and time-line

The following guidelines apply when filling in answers on an Excel spreadsheet.

- The Excel spreadsheet for a Pre-Practical or Practical are found on the AMS system. The spreadsheet contains unique parameter values for each student. Therefore, you must download the Excel spreadsheet from your own AMS account. Follow the instructions on the AMS spreadsheet carefully when filling in your answers.
- Final answers must be written as real numbers and not as fractions. Use at least 5 significant figures to write irrational numbers. Use decimal points and NOT decimal commas.
- Answers must include units where applicable. (Refer to the EECE Rules for Electronic Assessments of tests and examinations for guidelines on how to enter units. Also refer to the example list of units on the Excel spreadsheet.)

After filling in your answers an Excel spreadsheet for a Pre-Practical or Practical, you need to upload the Excel spreadsheet to AMS for that assessment to be marked. Each Pre-practical and Practical has a start time and an end-time as indicated on AMS. The following rules and time-line apply for all assessments for EBN 111 as illustrated by the figure below:

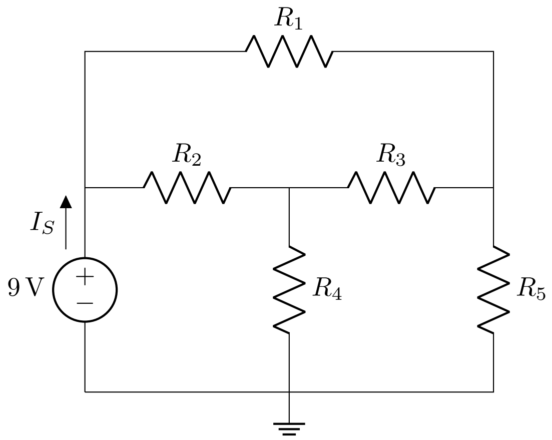
1. The Download Grace Period (DGP) is a 15 minute period before the official start of the assessment for students to download all documents related to the assessment and read all the necessary instructions.
2. The start of the green section after the DGP indicates the official start of the assessment.
3. The upward green arrow at the end of the green section indicates the official submission deadline for the assessment.
4. Students must aim to submit their Excel spreadsheet with their answers on the AMS before the official submission deadline.
5. Students can request assistance via **e-mail** ebn2023.practicals@gmail.com if they experience any technical issues with respect to uploading their Excel spreadsheets on the AMS.
6. An Upload Grace Period (UGP) of 15 minutes follows the official submission deadline. Students may submit their Excel spreadsheets with answers during this period without incurring any penalties. No email submissions will be accepted during the UGP.
7. A Penalty Period (PP) of 10 minutes follows the UGP. If a student submits an Excel spreadsheets during this period, the student incurs a 10% penalty to their final assessment mark.
8. No late submissions are accepted after the PP.




**By uploading an Excel spreadsheet to AMS, you declare the following:**

*The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, examinations and/or any other forms of assessment.*

## Section 1: Pre-Practical - Calculations and Simulation

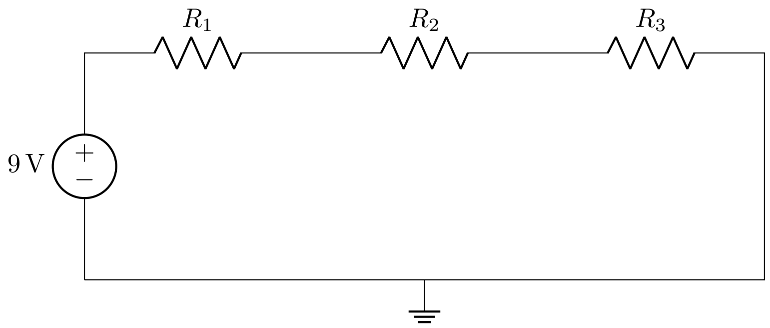
| **Questions 1 to 7 [8]** | |
|---|---|
| Given: $R_1$, $R_2$, $R_3$, $R_4$, $R_5$ | |
| **Circuit 1**  | |
| Complete the questions below through theoretical analysis of **Circuit 1**. | |
| **01** | Calculate the equivalent resistance $R_{eq}$ of the circuit as seen by the source. [1] |
| Build **Circuit 1** in LTspice and perform a **DC operating point** simulation. Complete the questions below. | |
| **02** | Measure $I_S$ in simulation. [1] |
| **03** | Measure the equivalent resistance $R_{eq}$ of the circuit as seen by the source in simulation. Hint: Use $I_S$ and Ohm's law. [1] |
| **04** | Measure the power dissipated by each resistor. Determine which one dissipates the most power.<br>Write only the resistor number as your answer. For example if $R_1$ dissipates the most power, write 1, if $R_2$ dissipates the most power, write 2, etc. If multiple resistors dissipate the same, maximum amount of power, provide only the first resistor number. For example, if $R_3$ and $R_5$ dissipate the same amount of maximum power, write 3. [2] |

Consider three resistors connected in series as shown in **Circuit 2**. Choose resistor values for $R_1$, $R_2$, and $R_3$ that will result in an equivalent resistance that is as close as possible to the theoretically calculated equivalent resistance $R_{eq}$ of **Circuit 1**.

**The chosen values have to be standard resistor values within the ranges specified below:**

$$\begin{aligned} 1\,\mathrm{k\Omega} &\leq R_1 < 10\,\mathrm{k\Omega} \\ 100\,\Omega &\leq R_2 < 1\,\mathrm{k\Omega} \\ 10\,\Omega &\leq R_3 < 100\,\Omega \end{aligned}$$

**Circuit 2**

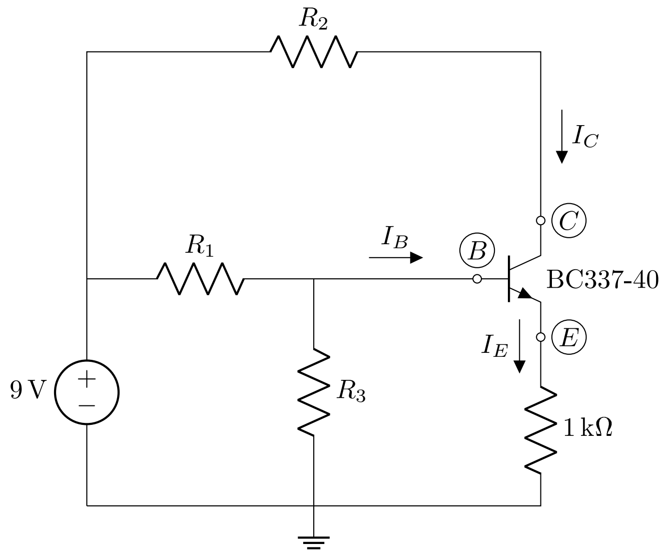



| | |
|---|---|
| **05** | Standard resistor chosen for $R_1$. [1] |
| **06** | Standard resistor chosen for $R_2$. [1] |
| **07** | Standard resistor chosen for $R_3$. [1] |

**Questions 8 to 23 [17]**

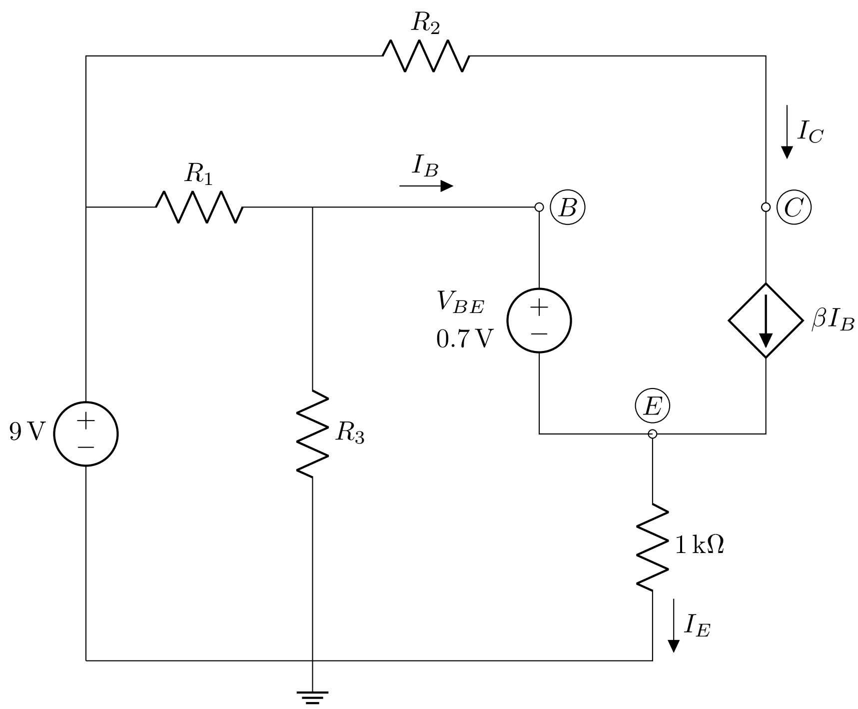
Given: $R_1$, $R_2$, $R_3$, $\beta_1$, $\beta_2$

**Circuit 3**




The transistor in **Circuit 3** can be modelled using the equivalent model of a transistor for theoretical calculations and simulations as shown in **Circuit 4**.

**Circuit 4**



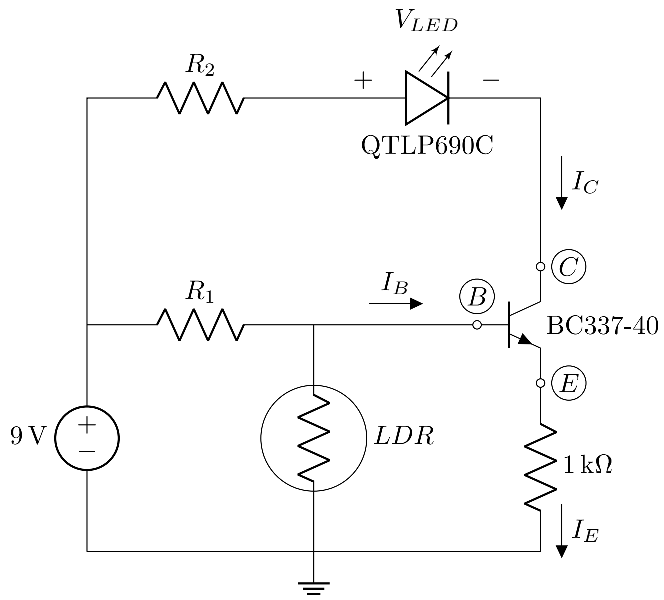

Complete the questions below through theoretical analysis of **Circuit 4** for the given $\beta$-values.

| | |
|---|---|
| **08** | Calculate $I_B$ when $\beta = \beta_1$. [1] |
| **09** | Calculate $I_C$ when $\beta = \beta_1$. [1] |
| **10** | Calculate $I_E$ when $\beta = \beta_1$. [1] |

| | |
|---|---|
| **11** | Calculate $I_B$ when $\beta = \beta_2$. [1] |
| **12** | Calculate $I_C$ when $\beta = \beta_2$. [1] |
| **13** | Calculate $I_E$ when $\beta = \beta_2$. [1] |
| Build **Circuit 4** in LTspice for the given $\beta$-values. Perform a **DC operating point** simulation and complete the questions below. | |
| **14** | Measure $I_B$ when $\beta = \beta_1$ in simulation. [1] |
| **15** | Measure $I_C$ when $\beta = \beta_1$ in simulation. [1] |
| **16** | Measure $I_E$ when $\beta = \beta_1$ in simulation. [1] |
| **17** | Measure $I_B$ when $\beta = \beta_2$ in simulation. [1] |
| **18** | Measure $I_C$ when $\beta = \beta_2$ in simulation. [1] |
| **19** | Measure $I_E$ when $\beta = \beta_2$ in simulation. [1] |
| Build **Circuit 3** in LTspice using the specified transistor **(BC337-40)**. Perform a **DC operating point** simulation and complete the questions below. | |
| **20** | Measure $I_B$ in simulation. [1] |
| **21** | Measure $I_C$ in simulation. [1] |
| **22** | Measure $I_E$ in simulation. [1] |
| **23** | Using the currents you measured, calculate the value of $\beta$ of the transistor you simulated. [2] |

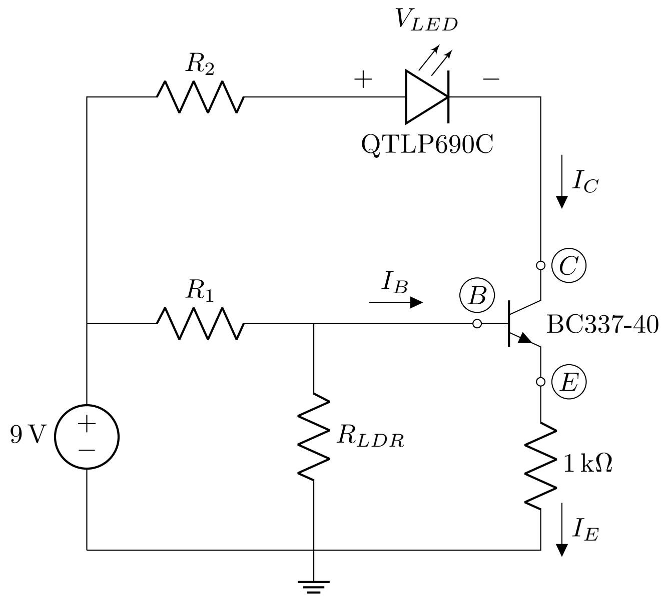
| **Questions 24 to 28 [7]** |
|---|
| Given: $R_1$, $R_2$, $R_{LDR}$ |

**Circuit 5**




The LDR in **Circuit 5** can be modelled as a resistor $R_{LDR}$ as shown in **Circuit 6**.

**Circuit 6**




Complete the questions below through theoretical analysis of **Circuit 6** assuming $V_{LED} = 2\,\text{V}$ and $V_{BE} = 0.7\,\text{V}$. Use the $\beta$ value you calculated in Q36.

| | |
|---|---|
| **24** | Calculate $I_{LED}$. [1] |
| **25** | Calculate the power $P_{LED}$ dissipated by the LED. [1] |

| | |
|---|---|
| Build **Circuit 6** in LTspice. Perform a **DC operating point** simulation and complete the questions below. Make sure to change the default LED to a **QTLP690C**. | |
| **26** | Measure $V_{LED}$ in simulation. [2] |
| **27** | Measure $I_{LED}$ in simulation. [2] |
| **28** | Measure the power $P_{LED}$ dissipated by the LED in simulation. [1] |

| **Questions 29 to 37 [18]** | |
|---|---|
| The resistance of the LDR in **Circuit 5** is dependent on the ambient light intensity. The circuit behavior at different light levels can be simulated by sweeping the resistance of resistor $R_{LDR}$ over a range that is typical for the LDR.<br>Build **Circuit 6** in LTspice. Perform a **DC sweep** simulation and sweep $R_{LDR}$ from $100\,\Omega$ to $5\,\mathrm{k}\Omega$ while measuring the current $I_{LED}$ and voltage $V_{LED}$. Complete the questions below using the graph you obtain in simulation. | |
| **29** | Measure the maximum current $I_{LEDmax}$ through the LED. [2] |
| **30** | Measure the voltage $V_{LEDmax}$ when the current $I_{LED}$ is maximum. [2] |
| **31** | Measure the maximum power $P_{LEDmax}$ dissipated by the LED. [2] |
| **32** | Measure the minimum current $I_{LEDmin}$ through the LED. [2] |
| **33** | Measure the voltage $V_{LEDmin}$ when the current $I_{LED}$ is minimum. [2] |
| **34** | Measure the minimum power $P_{LEDmin}$ dissipated by the LED in simulation. [2] |
| **35** | Measure the current $I_{LED}$ when $R_{LDR} = 2.5\,\mathrm{k}\Omega$. [2] |
| **36** | Measure the voltage $V_{LED}$ when $R_{LDR} = 2.5\,\mathrm{k}\Omega$. [2] |
| **37** | Measure the power $P_{LED}$ dissipated by the LED when $R_{LDR} = 2.5\,\mathrm{k}\Omega$ in simulation. [2] |

**Total Section 1: [50]**

## Section 2: Practical Demonstration

**Question 38 [20]**

Given: $R_1$, $R_2$

The values for $R_1$ and $R_2$ should be same as used for Q24 to Q28 above.

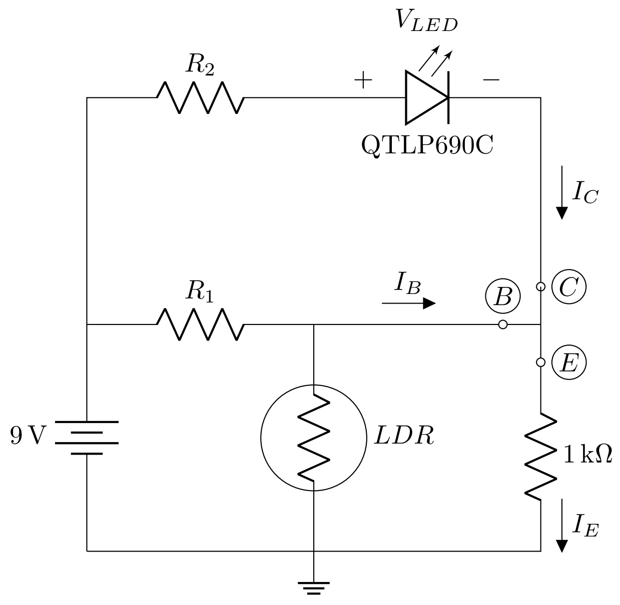
Build **Circuit 8** shown below on a breadboard using the components provided in the component kit. A 9 V battery can be used to power the circuit. This circuit will be used for the practical session. It is advised to read through the next session and build the circuit in such a way that the changes required for the Practical will be easy to implement.

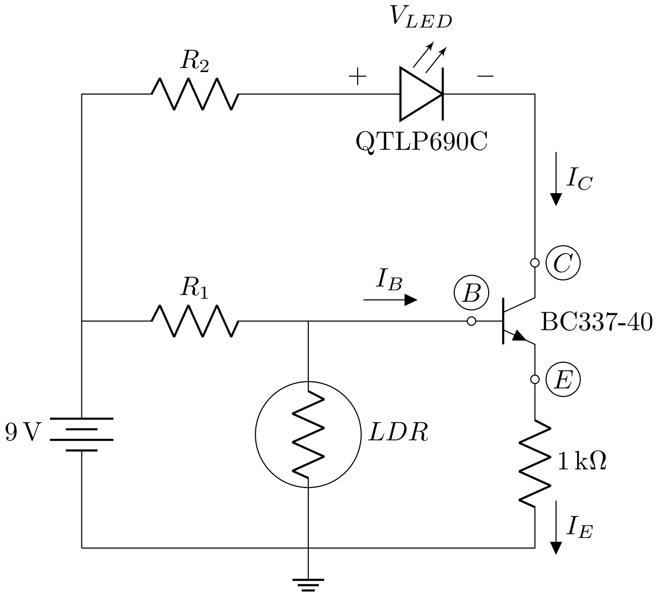
After building the circuit you can test it by placing the circuit in a dark or brightly lit room. The brightness of the LED should change depending on the brightness of the light falling on the LDR. The BC337-40 transistors are sensitive components and can be blown if too much voltage is applied to them. They are not as easy as an LED to diagnose if they are blown, hence it is recommended that **Circuit 7** is built first to check that the circuit is correct (The LED should dim and brighten as in practical 1), before adding the transistor as in **Circuit 8**.

When entering the lab for your allocated practical session, the circuit will be evaluated by an Assistant Lecturer. The circuit will be evaluated according to:

- Fully functioning circuit presented on arrival at the practical venue [20]
- Non-working circuit presented upon entry and fixed during the session [15]
- No circuit is presented at entry but a functioning circuit is built and presented during the session [10]
- No functioning circuit presented before the end of the session [0]

**Circuit 7**



| **Circuit 8** | |
|---|---|
|  | |
| **38** | Circuit evaluation for **Circuit 8**. [20] |

| **Total Section 2: [20]** |
|---|

## Section 3: Practical Measurements

| Question 39 [5] | |
|---|---|
| Set up the laboratory power supply to supply 9 V DC. Use the multimeter to measure the voltage $V_S$ supplied by the power supply and verify that it is between 8.9 V and 9.1 V.<br><br>**The measured supply voltage $V_S$ will be used in the marking of subsequent questions. Take care to ensure that this step is done correctly.**<br><br>**Once the supply voltage is set, it should not be changed under any circumstances!** | |
| **39** | Enter the measured supply voltage $V_S$. [5] |

**Questions 40 to 52 [54]**

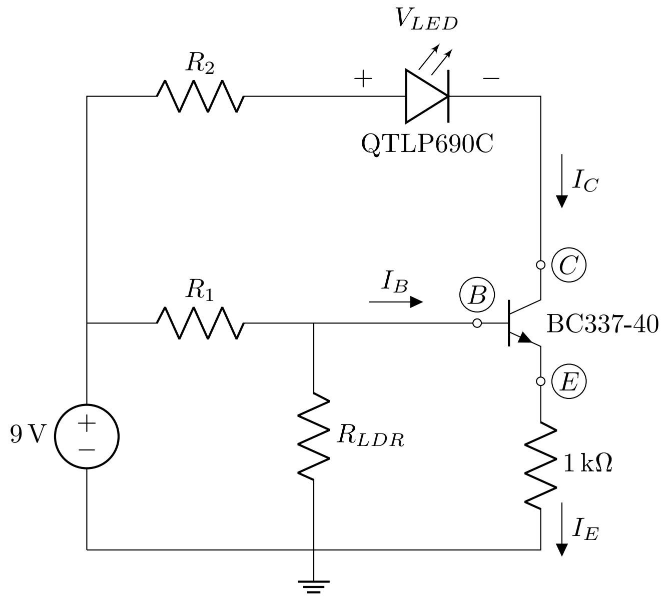
Given: $R_1$, $R_2$, $R_{LDR}$

**Important:** The values for $R_1$, $R_2$ and $R_{LDR}$ are the same as for Q41 to Q54 above and will be used throughout the rest of the Practical.

Replace the LDR in **Circuit 8** (the circuit that was built for the demonstration) with a 5 kΩ potentiometer as shown in **Circuit 9** below.

Use the multimeter to measure the voltages/currents specified below. Remember that current must be measured in series. However, current can be calculated by measuring the voltage across a component first and then using Ohm's law.

**Circuit 9**




Turn the potentiometer to where the LED is brightest.

| | |
|---|---|
| **40** | Measure $V_{LEDmax}$. [6] |
| **41** | Measure $V_{R2}$. [6] |
| **42** | Measure $I_{LEDmax}$. [2] |
| **43** | Measure the power $P_{LEDmax}$ dissipated by the LED. [2] |

Turn the potentiometer to where the LED is dimmest.

| | |
|---|---|
| **44** | Measure $V_{LEDmin}$ [6] |
| **45** | Measure $V_{R2}$. [6] |
| **46** | Measure $I_{LEDmin}$ [2] |
| **47** | Measure the power $P_{LEDmin}$ dissipated by the LED. [2] |

Turn the shaft/wiper of the potentiometer to approximately the middle of its range. Try to do this as accurately as possible by first finding both extremes of the potentiometer and then moving the shaft/wiper to a position between the extremes.

| | |
|---|---|
| **48** | Measure $V_{LED}$ [6] |
| **49** | Measure $V_{R2}$. [6] |
| **50** | Measure $I_{LED}$ [2] |

| | |
|---|---|
| **51** | Measure the power $P_{LED}$ dissipated by the LED. [2] |

Take the potentiometer out of the circuit and measure its resistance $R_{pot}$ when the wiper/shaft is in the middle of the range.

**Be careful to not change the wiper/shaft position**. Check how it was connected in the circuit and make sure the resistance is measured between the correct legs. Since the human body is also a resistor, take care not to touch the probes or component legs while making resistance measurements.

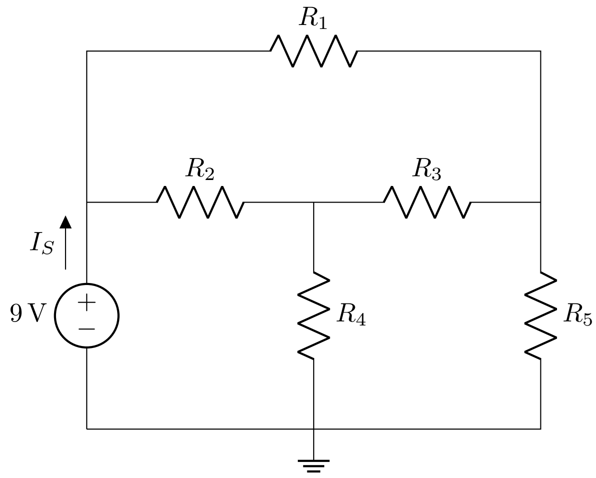
| | |
|---|---|
| **52** | Measure $R_{pot}$ [6] |

**Questions 53 to 57 [21]**

Given: $R_1$, $R_2$, $R_3$, $R_4$, $R_5$

**Important:** The values for $R_1$, $R_2$, $R_3$, $R_4$, and $R_5$ are the same as for Q1 to Q7 above.

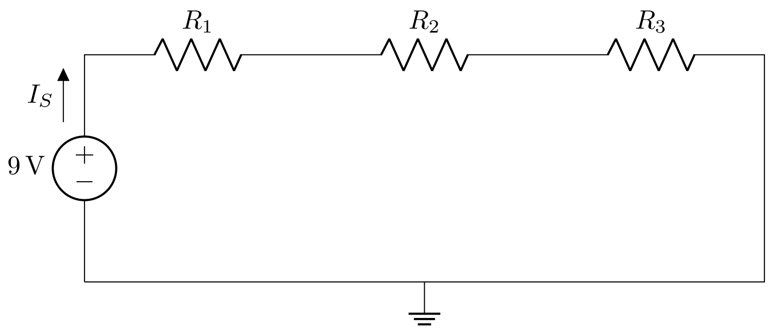
**Circuit 10**




| | |
|---|---|
| **53** | Measure the equivalent resistance $R_{eq}$ of the circuit as seen by the source. [6] |
| **54** | Measure $I_S$. [2] |
| **55** | Measure the power dissipated by each resistor and establish which one dissipates the most power.<br>Write only the resistor number as your answer. For example if $R_1$ dissipates the most power, write 1, if $R_2$ dissipates the most power, write 2, etc. If multiple resistors dissipate the same, maximum amount of power, provide only the first resistor number. For example, if $R_3$ and $R_5$ dissipate the same amount of maximum power, write 3. [5] |

Consider three resistors connected in series as shown in **Circuit 11**. Build the circuit using the resistor values that were chosen in Section 1 for **Circuit 2** to have a similar equivalent resistance as **Circuit 10**.

**Circuit 11**




| | |
|---|---|
| **56** | Measure the equivalent resistance $R_{eq}$ of the circuit as seen by the source. [6] |
| **57** | Measure $I_S$. [2] |

**Total Section 3: [80]**

**Grand Total : [150]**